



# **SSD222D**

## **Smart Display CAM Controller with Embedded 16-bit 128MB DDR3**

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**Preliminary Product Brief Version 0.6**

## REVISION HISTORY

| Revision No. | Description                                                                                                                                                                                                                               | Date       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 0.1          | <ul style="list-style-type: none"> <li>Initial release</li> </ul>                                                                                                                                                                         | 09/11/2020 |
| 0.2          | <ul style="list-style-type: none"> <li>Updated Signal Description by adding Power Domain Information</li> <li>Updated Ambient Operation Temperature Range</li> <li>Added Minimum Order Quantity and Moisture Sensitivity Level</li> </ul> | 01/21/2021 |
| 0.3          | <ul style="list-style-type: none"> <li>Added Hardware Power Sequence Procedure, MSPI Operation Example, and Thermal Resistance</li> </ul>                                                                                                 | 02/26/2021 |
| 0.4          | <ul style="list-style-type: none"> <li>Updated Features</li> </ul>                                                                                                                                                                        | 05/31/2021 |
| 0.5          | <ul style="list-style-type: none"> <li>Added more information</li> </ul>                                                                                                                                                                  | 10/14/2021 |
| 0.6          | <ul style="list-style-type: none"> <li>Updated DRAM related description</li> </ul>                                                                                                                                                        | 02/22/2022 |

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## TABLE OF CONTENTS

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>TABLE OF CONTENTS</b>                         | <b>ii</b> |
| <b>1. FEATURES</b>                               | <b>1</b>  |
| <b>2. BLOCK DIAGRAM</b>                          | <b>3</b>  |
| <b>3. GENERAL DESCRIPTION</b>                    | <b>4</b>  |
| <b>4. PACKAGE INFORMATION</b>                    | <b>5</b>  |
| 4.1. Pin Diagram                                 | 5         |
| 4.2. Signal Description                          | 6         |
| 4.3. Mechanical Dimensions                       | 12        |
| 4.4. Ordering Guide                              | 13        |
| 4.5. Marking Information                         | 13        |
| <b>5. ELECTRICAL SPECIFICATIONS</b>              | <b>14</b> |
| 5.1. Interface Characteristics                   | 14        |
| 5.2. Recommended Operating Conditions            | 15        |
| 5.3. Absolute Maximum Ratings                    | 15        |
| <b>6. HARDWARE POWER SEQUENCE PROCEDURE</b>      | <b>16</b> |
| 6.1. Hardware Reset                              | 16        |
| 6.2. External Reset IC with External LDO         | 17        |
| 6.3. Without External Reset IC with External LDO | 18        |
| <b>7. MSPI OPERATION EXAMPLE</b>                 | <b>19</b> |
| 7.1. Set MIIC Speed                              | 19        |
| <b>8. PSPI INTERFACE</b>                         | <b>21</b> |
| <b>9. EPHY INTERFACE</b>                         | <b>22</b> |
| 9.1. 100BASE-TX                                  | 22        |
| 9.2. 10BASE-T                                    | 22        |
| <b>10. THERMAL RESISTANCE (°C/W)</b>             | <b>23</b> |
| 10.1. Thermal simulation mode                    | 23        |

## 1. FEATURES

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### ■ High Performance Processor Core

- ARM Cortex-A7 Dual Core up to 1 GHz
- 16KB I-Cache/16KB D-Cache/128KB L2-Cache
- Neon and FPU
- Memory Management Unit for Linux support
- DMA Engine

### ■ Image/Video Processor

- Supports 8/10-bit parallel interface for raw data input
- Supports max. two MIPI interfaces with 2 or 1 data lane and 2 clock lanes, up to 1.5GHz
- Supports 8/10-bit BT.601/656 parallel interface
- ISP processing performance up to 1920x1080p30
- Bad pixel compensation
- Temporal-domain Noise Reduction (3DNR)
- Bayer domain Spatial-domain Noise Reduction (2DNR)
- Bayer domain filter to remove purple false color in highlight regions
- Optical black correction
- Lens shading compensation
- Auto White Balance (AWB) / Auto Exposure (AE) / Auto Focus (AF)
- CFA color interpolation
- Color correction
- Gamma correction
- Video stabilization
- Frame buffer data compression and de-compression to save memory bandwidth
- Wide Dynamic Range (WDR) with local tone mapping

### ■ JPEG Encoder

- Supports JPEG baseline encoding
- Supports YUV422 or YUV420 formats

- Supports max. resolution 720p (1280x720) with 30fps

### ■ Display Subsystem

- Built-in contrast, brightness, sharpness, and saturation, 3D NR, Gamma control
- TTL output up to 1280x800 60fps with RGB565 or RGB666 or RGB888 format
- BT.656 output up to 720p60
- Serial RGB up to 800x600 60fps
- Supports SPI panel, clock frequency up to 54MHz
- Supports FHD graphic layer with Index 4/8, ARGB1555/ARGB4444/ARGB8888, and RGB565 format
- Supports UI/OSD layer with max. resolution 1280x800

### ■ 2D Graphics Engine

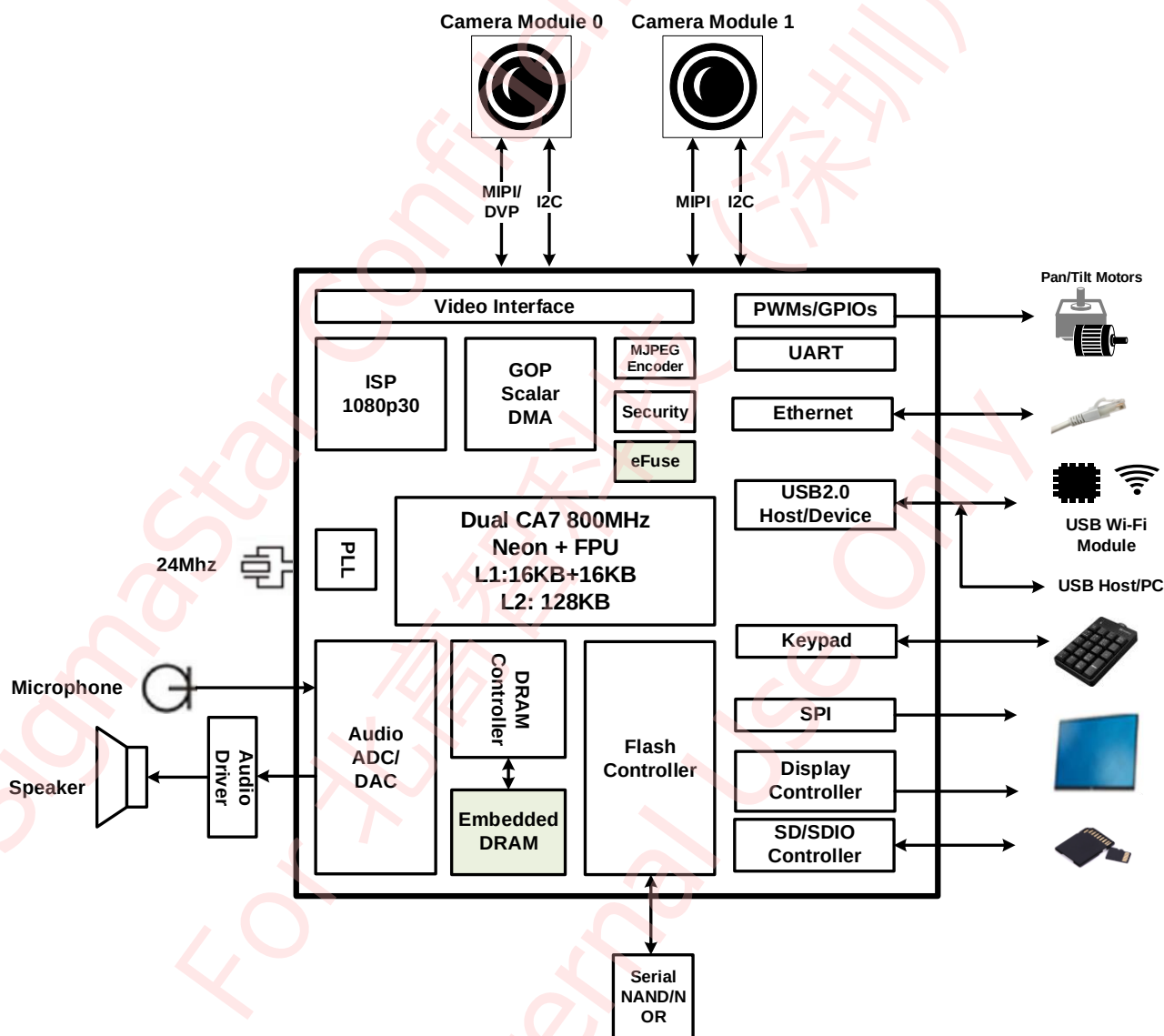
- Line draw
- Rectangle/gradient rectangle fill
- Bitblt/Stretch Bitblt/Italic Bitblt
- Palette mode (1/2/4/8-bit)
- Format transformation
- Color space conversion
- Clipping
- Alpha blending
- Rotation/Mirror
- Dither

### ■ Audio Processor

- Three mono ADCs or one mono + one stereo ADC for microphone input
- Two stereo DMIC inputs
- I2S TDM 8-channel, RX 2/4/8 channels, TX 2 channels
- One stereo DAC for lineout
- One HP Driver headphone set
- I2S supports 8K/16K/32K/48K/96KHz sampling rate audio recording

- ADC Pre-Amp gain supports 0dB, 6dB, 13dB, 23dB, 30dB, and 36dB
- ADC boost gain supports -6dB ~ 15dB or 0dB ~ 21dB with interval 3dB
- ADC digital gain supports -63.5dB ~ 33dB with interval 0.5dB, can be muted to zero
- SNR of DR A-Weighted ADC > 90dB (@gain = 0dB)
- **NOR/NAND Flash Interface**
  - Supports 1/2/4-bit SPI-NOR / NAND flash with two chip selects
- **SDIO 2.0 Interface**
  - Compatible with SDIO spec. 2.0, data bus 1/4 bit mode
  - Compatible with SD spec. 2.0, data bus 1/4 bit mode
- **USB 2.0 Interface**
  - One USB2.0 configurable host and device
    - Host mode supports EHCI specification
    - Device mode supports 4 end points
- **DRAM Memory**
  - Embedded 16-bit 128MB DDR3 memory with max. 1866Mbps
  - Supports auto-refresh and self-refresh mode
- **Ethernet**
  - Supports one Ethernet port
  - Supports 10/100Mbps half/full-duplex
  - One built-in 10/100M Ethernet PHY
  - Supports one RMI to connect external PHY
  - Supports two LEDs for ePHY
- **Security Engines**
  - Supports AES/DES/3DES/RSA/SHA-1/SHA-256
  - Supports secure booting
- **Boot options**
  - ROM
  - SPI NOR
  - SPI NAND with ECC
  - SD Card and USB
- **Peripherals**
  - Dedicated GPIOs for system control
  - Four PWM outputs
  - Three generic UARTs and one fast UART with flow control
  - Three generic timers and one watchdog timer
  - Two SPI masters
  - Two I2C masters
  - Keypad supports up to 7x7, single mode
- **Miscellaneous**
  - Built-in eFuse with 1024-bit to store device ID, AES key, chip configurations, etc.
  - Built-in power on reset (POR)
  - Built-in SAR ADC with 2-channel analog inputs for different kinds of applications
- **Operating Voltage Range**
  - Core: 0.9V
  - I/O: 1.8V/3.3V
  - DRAM: 1.5V (DDR3)
  - Power Consumption: TBD.
  - Ambient Temperature -20°C ~ 85°C
- **Package**
  - 128-pin QFN, 12.3mm x 12.3mm
  - Moisture Sensitivity Level (MSL): 3

## 2. BLOCK DIAGRAM



### 3. GENERAL DESCRIPTION

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The SSD222D is a highly integrated SOC product for face access and smart display applications.

Based on ARM Cortex-A7 dual-core, the SSD222D integrates image sensor interface, advanced ISP, high performance JPEG encoder, 2D graphics engine, TTL/serial RGB display with adjustable picture quality engine and other useful peripherals.

A typical utilization of the SSD222D application processor is demonstrated in the block diagram. The completed system includes NOR/NAND flash, DRAM, SD card, and USB port, and diversified audio connection. Before output to the panel, the images can be enhanced with respect to brightness/contrast/saturation/sharpness to give the best picture quality.

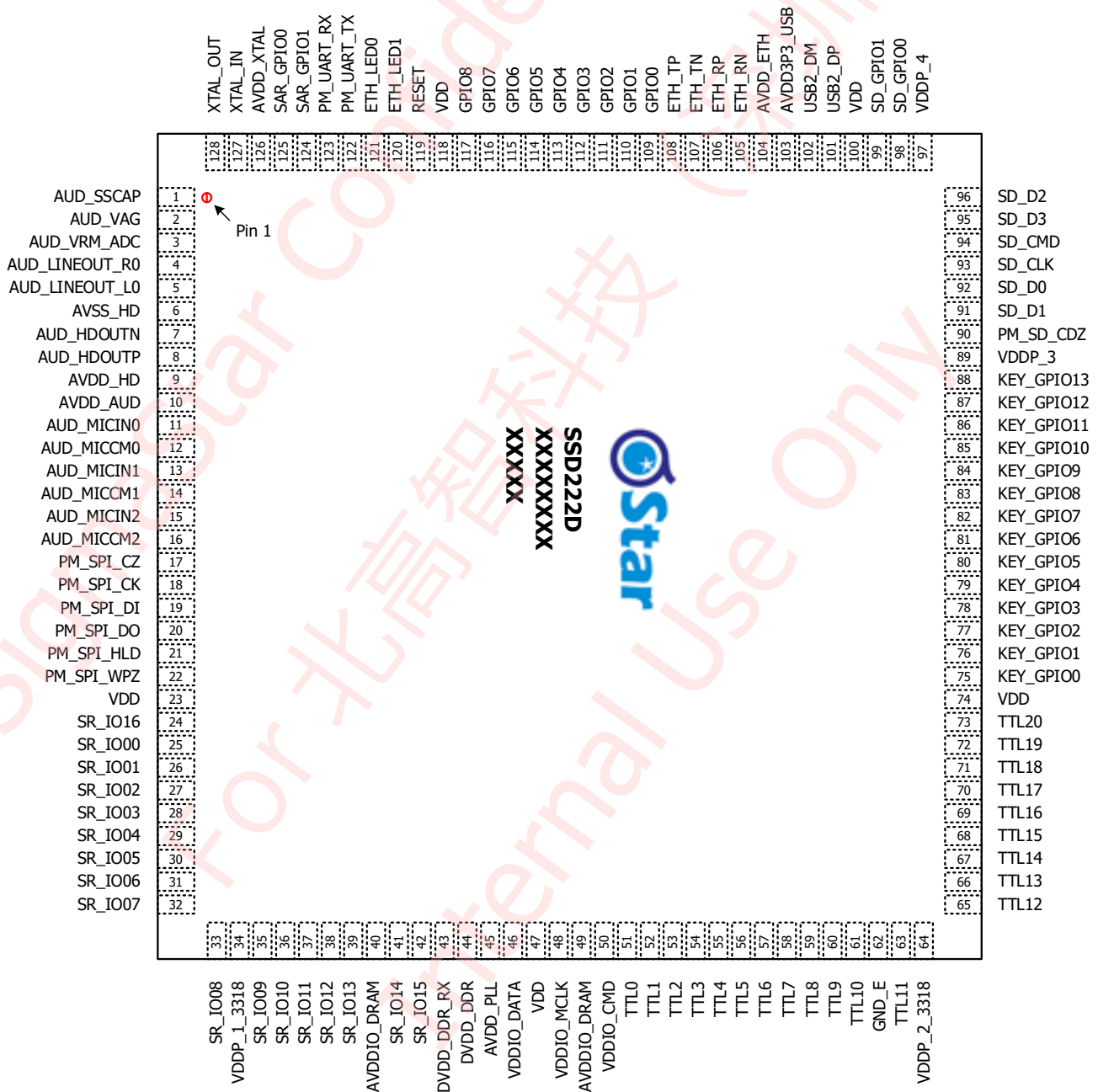
The NOR or NAND flash is usually reserved for operating system and application software. Moreover, other peripherals like SAR ADC, Audio ADC/DAC, UARTs, PWMs, GPIOs and SPI are supported to realize applications with maximal flexibility.

The SSD222D supports secure booting and personalization authentication mechanism for securing system. The AES/DES/3DES cipher engines could also help encrypt the compressed video/audio streams for privacy protection.

The SSD222D, powered by SigmaStar Technology, comes with a complete hardware platform and software SDK, allowing customers to speed up "Time-to-Market."

## 4. PACKAGE INFORMATION

### 4.1. Pin Diagram



## 4.2. Signal Description

| Signal Name                   | Signal Type | Function                                                                      | Power Domain | Pin Location |
|-------------------------------|-------------|-------------------------------------------------------------------------------|--------------|--------------|
| <b>System Reset Interface</b> |             |                                                                               |              |              |
| RESET                         | I           | System Reset (Active High)                                                    | AVDD_XTAL    | 119          |
| <b>Debug UART Interface</b>   |             |                                                                               |              |              |
| PM_UART_RX                    | I           | Debug UART Receive Data Input with Pull Up Resistor / Slave I2C Serial Clock  | AVDD_XTAL    | 123          |
| PM_UART_TX                    | O           | Debug UART Transmit Data Output with Pull Up Resistor / Slave I2C Serial Data | AVDD_XTAL    | 122          |
| <b>System Interface</b>       |             |                                                                               |              |              |
| XTAL_IN                       | I           | 24MHz Crystal Input                                                           | AVDD_XTAL    | 127          |
| XTAL_OUT                      | O           | 24MHz Crystal Output                                                          | AVDD_XTAL    | 128          |
| <b>SPI Flash Interface</b>    |             |                                                                               |              |              |
| PM_SPI_CZ                     | O           | SPI Chip Select (Active Low)                                                  | AVDD_XTAL    | 17           |
| PM_SPI_CK                     | O           | SPI Clock                                                                     | AVDD_XTAL    | 18           |
| PM_SPI_DI                     | O           | SPI Serial Data To Device (MOSI)                                              | AVDD_XTAL    | 19           |
| PM_SPI_DO                     | I           | SPI Serial Data From Device (MISO)                                            | AVDD_XTAL    | 20           |
| PM_SPI_WPZ                    | O           | SPI Write Protect                                                             | AVDD_XTAL    | 22           |
| PM_SPI_HLD                    | O           | SPI Hold                                                                      | AVDD_XTAL    | 21           |
| <b>GPIO Interface</b>         |             |                                                                               |              |              |
| GPIO0                         | I/O         | General Purpose Input/Output 0                                                | AVDD_XTAL    | 109          |
| GPIO1                         | I/O         | General Purpose Input/Output 1                                                | AVDD_XTAL    | 110          |
| GPIO2                         | I/O         | General Purpose Input/Output 2                                                | AVDD_XTAL    | 111          |
| GPIO3                         | I/O         | General Purpose Input/Output 3                                                | AVDD_XTAL    | 112          |
| GPIO4                         | I/O         | General Purpose Input/Output 4                                                | AVDD_XTAL    | 113          |
| GPIO5                         | I/O         | General Purpose Input/Output 5                                                | AVDD_XTAL    | 114          |
| GPIO6                         | I/O         | General Purpose Input/Output 6                                                | AVDD_XTAL    | 115          |
| GPIO7                         | I/O         | General Purpose Input/Output 7                                                | AVDD_XTAL    | 116          |
| GPIO8                         | I/O         | General Purpose Input/Output 8                                                | AVDD_XTAL    | 117          |
| <b>KEY GPIO Interface</b>     |             |                                                                               |              |              |
| KEY_GPIO0                     | I/O         | KEY General Purpose Input/Output 0                                            | VDDP_3       | 75           |
| KEY_GPIO1                     | I/O         | KEY General Purpose Input/Output 1                                            | VDDP_3       | 76           |

| Signal Name                   | Signal Type | Function                                                        | Power Domain | Pin Location |
|-------------------------------|-------------|-----------------------------------------------------------------|--------------|--------------|
| KEY_GPIO2                     | I/O         | KEY General Purpose Input/Output 2                              | VDDP_3       | 77           |
| KEY_GPIO3                     | I/O         | KEY General Purpose Input/Output 3                              | VDDP_3       | 78           |
| KEY_GPIO4                     | I/O         | KEY General Purpose Input/Output 4                              | VDDP_3       | 79           |
| KEY_GPIO5                     | I/O         | KEY General Purpose Input/Output 5                              | VDDP_3       | 80           |
| KEY_GPIO6                     | I/O         | KEY General Purpose Input/Output 6                              | VDDP_3       | 81           |
| KEY_GPIO7                     | I/O         | KEY General Purpose Input/Output 7                              | VDDP_3       | 82           |
| KEY_GPIO8                     | I/O         | KEY General Purpose Input/Output 8                              | VDDP_3       | 83           |
| KEY_GPIO9                     | I/O         | KEY General Purpose Input/Output 9                              | VDDP_3       | 84           |
| KEY_GPIO10                    | I/O         | KEY General Purpose Input/Output 10                             | VDDP_3       | 85           |
| KEY_GPIO11                    | I/O         | KEY General Purpose Input/Output 11                             | VDDP_3       | 86           |
| KEY_GPIO12                    | I/O         | KEY General Purpose Input/Output 12                             | VDDP_3       | 87           |
| KEY_GPIO13                    | I/O         | KEY General Purpose Input/Output 13                             | VDDP_3       | 88           |
| <b>SAR ADC Interface</b>      |             |                                                                 |              |              |
| SAR_GPIO0                     | I           | General Purpose Input/Output or Muxed to SARADC Input Channel 0 | AVDD_XTAL    | 125          |
| SAR_GPIO1                     | I           | General Purpose Input/Output or Muxed to SARADC Input Channel 1 | AVDD_XTAL    | 124          |
| <b>Image Sensor Interface</b> |             |                                                                 |              |              |
| SR_IO00                       | I/O         | Sensor General Purpose Input/Output 0                           | VDDP_1_3318  | 25           |
| SR_IO01                       | I/O         | Sensor General Purpose Input/Output 1                           | VDDP_1_3318  | 26           |
| SR_IO02                       | I/O         | Sensor General Purpose Input/Output 2                           | VDDP_1_3318  | 27           |
| SR_IO03                       | I/O         | Sensor General Purpose Input/Output 3                           | VDDP_1_3318  | 28           |
| SR_IO04                       | I/O         | Sensor General Purpose Input/Output 4                           | VDDP_1_3318  | 29           |
| SR_IO05                       | I/O         | Sensor General Purpose Input/Output 5                           | VDDP_1_3318  | 30           |
| SR_IO06                       | I/O         | Sensor General Purpose Input/Output 6                           | VDDP_1_3318  | 31           |
| SR_IO07                       | I/O         | Sensor General Purpose Input/Output 7                           | VDDP_1_3318  | 32           |
| SR_IO08                       | I/O         | Sensor General Purpose Input/Output 8                           | VDDP_1_3318  | 33           |
| SR_IO09                       | I/O         | Sensor General Purpose Input/Output 9                           | VDDP_1_3318  | 35           |
| SR_IO10                       | I/O         | Sensor General Purpose Input/Output 10                          | VDDP_1_3318  | 36           |
| SR_IO11                       | I/O         | Sensor General Purpose Input/Output 11                          | VDDP_1_3318  | 37           |
| SR_IO12                       | I/O         | Sensor General Purpose Input/Output 12                          | VDDP_1_3318  | 38           |
| SR_IO13                       | I/O         | Sensor General Purpose Input/Output 13                          | VDDP_1_3318  | 39           |
| SR_IO14                       | I/O         | Sensor General Purpose Input/Output 14                          | VDDP_1_3318  | 41           |
| SR_IO15                       | I/O         | Sensor General Purpose Input/Output 15                          | VDDP_1_3318  | 42           |

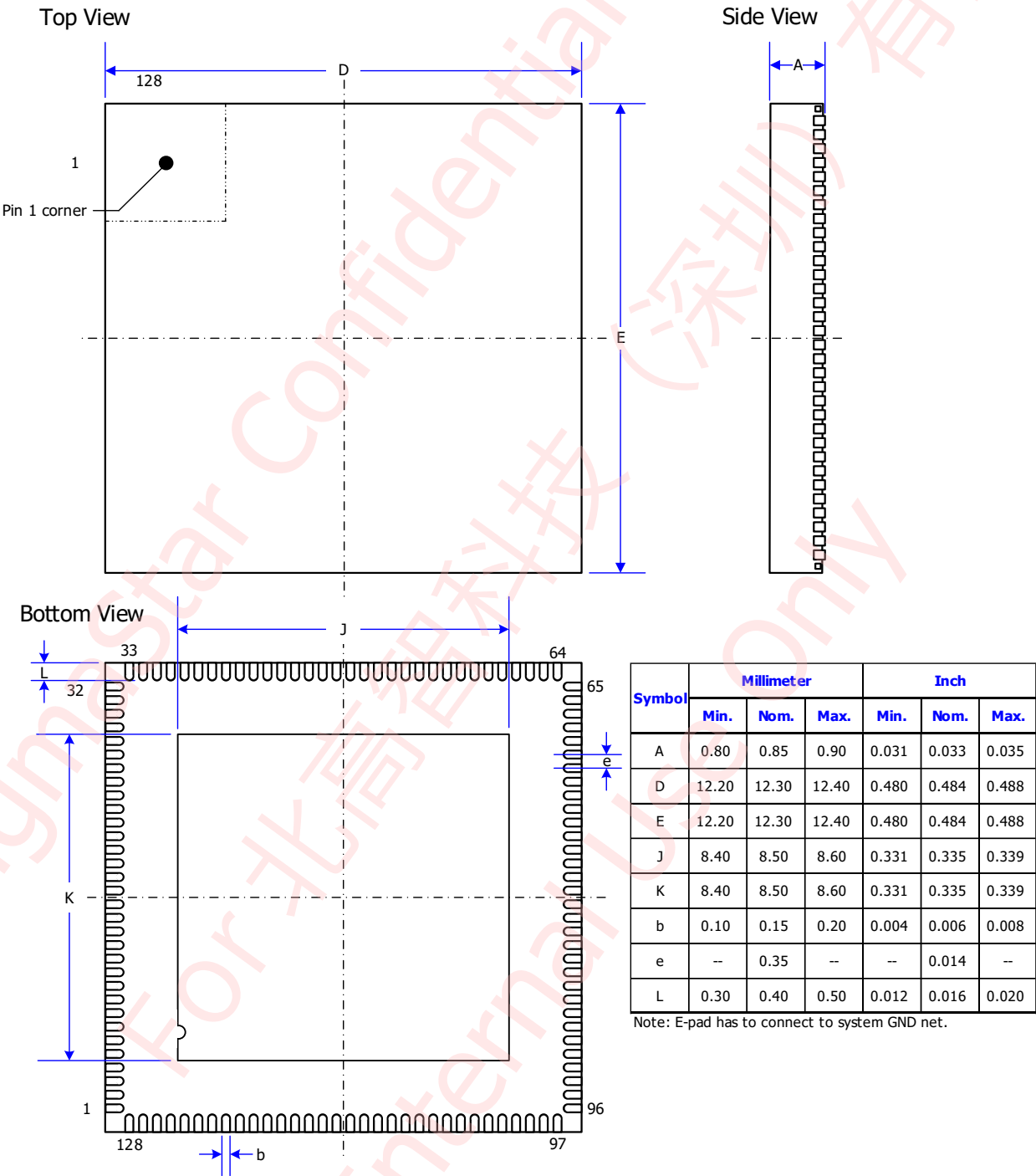
| Signal Name          | Signal Type | Function                                     | Power Domain | Pin Location |
|----------------------|-------------|----------------------------------------------|--------------|--------------|
| SR_IO16              | I/O         | Sensor General Purpose Input/Output 16       | VDDP_1_3318  | 24           |
| <b>TTL Interface</b> |             |                                              |              |              |
| TTL0                 | I/O         | Parallel LCD General Purpose Input/Output 0  | VDDP_2_3318  | 51           |
| TTL1                 | I/O         | Parallel LCD General Purpose Input/Output 1  | VDDP_2_3318  | 52           |
| TTL2                 | I/O         | Parallel LCD General Purpose Input/Output 2  | VDDP_2_3318  | 53           |
| TTL3                 | I/O         | Parallel LCD General Purpose Input/Output 3  | VDDP_2_3318  | 54           |
| TTL4                 | I/O         | Parallel LCD General Purpose Input/Output 4  | VDDP_2_3318  | 55           |
| TTL5                 | I/O         | Parallel LCD General Purpose Input/Output 5  | VDDP_2_3318  | 56           |
| TTL6                 | I/O         | Parallel LCD General Purpose Input/Output 6  | VDDP_2_3318  | 57           |
| TTL7                 | I/O         | Parallel LCD General Purpose Input/Output 7  | VDDP_2_3318  | 58           |
| TTL8                 | I/O         | Parallel LCD General Purpose Input/Output 8  | VDDP_2_3318  | 59           |
| TTL9                 | I/O         | Parallel LCD General Purpose Input/Output 9  | VDDP_2_3318  | 60           |
| TTL10                | I/O         | Parallel LCD General Purpose Input/Output 10 | VDDP_2_3318  | 61           |
| TTL11                | I/O         | Parallel LCD General Purpose Input/Output 11 | VDDP_2_3318  | 63           |
| TTL12                | I/O         | Parallel LCD General Purpose Input/Output 12 | VDDP_2_3318  | 65           |
| TTL13                | I/O         | Parallel LCD General Purpose Input/Output 13 | VDDP_2_3318  | 66           |
| TTL14                | I/O         | Parallel LCD General Purpose Input/Output 14 | VDDP_2_3318  | 67           |
| TTL15                | I/O         | Parallel LCD General Purpose Input/Output 15 | VDDP_2_3318  | 68           |
| TTL16                | I/O         | Parallel LCD General Purpose Input/Output 16 | VDDP_2_3318  | 69           |
| TTL17                | I/O         | Parallel LCD General Purpose Input/Output 17 | VDDP_2_3318  | 70           |

| Signal Name                       | Signal Type | Function                                                                                                                | Power Domain | Pin Location |
|-----------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| TTL18                             | I/O         | Parallel LCD General Purpose Input/Output 18                                                                            | VDDP_2_3318  | 71           |
| TTL19                             | I/O         | Parallel LCD General Purpose Input/Output 19                                                                            | VDDP_2_3318  | 72           |
| TTL20                             | I/O         | Parallel LCD General Purpose Input/Output 20                                                                            | VDDP_2_3318  | 73           |
| <b>10/100M Ethernet Interface</b> |             |                                                                                                                         |              |              |
| ETH_RN                            | I           | 10/100M Ethernet Differential Pair of Receiver Signal Negative                                                          | AVDD_ETH     | 105          |
| ETH_RP                            | I           | 10/100M Ethernet Differential Pair of Receiver Signal Positive                                                          | AVDD_ETH     | 106          |
| ETH_TN                            | O           | 10/100M Ethernet Differential Pair of Transmitter Signal Negative                                                       | AVDD_ETH     | 107          |
| ETH_TP                            | O           | 10/100M Ethernet Differential Pair of Transmitter Signal Positive                                                       | AVDD_ETH     | 108          |
| ETH_LED0                          | O           | 10/100M Ethernet LED0 Control Driven Active When Linked                                                                 | AVDD_XTAL    | 121          |
| ETH_LED1                          | O           | 10/100M Ethernet LED1 Control Driven Active When Linked in 100 Base-TX and Blinking When Transmitting or Receiving Data | AVDD_XTAL    | 120          |
| <b>SD 2.0 Card Interface</b>      |             |                                                                                                                         |              |              |
| SD_CLK                            | O           | SD 2.0 Clock                                                                                                            | VDDP_4       | 93           |
| SD_CMD                            | O           | SD 2.0 Command                                                                                                          | VDDP_4       | 94           |
| SD_D0                             | I/O         | SD 2.0 Data Bus 0                                                                                                       | VDDP_4       | 92           |
| SD_D1                             | I/O         | SD 2.0 Data Bus 1                                                                                                       | VDDP_4       | 91           |
| SD_D2                             | I/O         | SD 2.0 Data Bus 2                                                                                                       | VDDP_4       | 96           |
| SD_D3                             | I/O         | SD 2.0 Data Bus 3                                                                                                       | VDDP_4       | 95           |
| PM_SD_CDZ                         | I           | Power Manage SD 2.0 Card Detect                                                                                         | VDDP_4       | 90           |
| SD_GPIO0                          | I/O         | SD General Purpose Input/Output 0                                                                                       | VDDP_4       | 98           |
| SD_GPIO1                          | I/O         | SD General Purpose Input/Output 1                                                                                       | VDDP_4       | 99           |
| <b>Audio Line Out Interface</b>   |             |                                                                                                                         |              |              |
| AUD_LINEOUT_L0                    | O           | Audio Left Channel Line Output                                                                                          | AVDD_AUD     | 5            |
| AUD_LINEOUT_R0                    | O           | Audio Right Channel Line Output                                                                                         | AVDD_AUD     | 4            |
| AUD_VAG                           | O           | Audio Reference Voltage from 1/2 AVDD_AUD                                                                               | AVDD_AUD     | 2            |

| Signal Name                           | Signal Type     | Function                                         | Power Domain | Pin Location         |
|---------------------------------------|-----------------|--------------------------------------------------|--------------|----------------------|
| AUD_VRM_ADC                           | I               | Audio Reference Voltage for ADC                  | AVDD_AUD     | 3                    |
| AUD_SSCAP                             | I               | Audio Reference CAP                              | AVDD_AUD     | 1                    |
| <b>Audio Head Phone Out Interface</b> |                 |                                                  |              |                      |
| AUD_HDOUTN                            | O               | Audio Head Phone Out Differential Pair, Negative | AVDD_HD      | 7                    |
| AUD_HDOUTP                            | O               | Audio Head Phone Out Differential Pair, Positive | AVDD_HD      | 8                    |
| <b>Analog Microphone Interface</b>    |                 |                                                  |              |                      |
| AUD_MICIN0                            | I               | Audio Left Channel Microphone Positive Input     | AVDD_AUD     | 11                   |
| AUD_MICCM0                            | I               | Audio Left Channel Microphone Negative Input     | AVDD_AUD     | 12                   |
| AUD_MICIN1                            | I               | Audio Right Channel Microphone Positive Input    | AVDD_AUD     | 13                   |
| AUD_MICCM1                            | I               | Audio Right Channel Microphone Negative Input    | AVDD_AUD     | 14                   |
| AUD_MICIN2                            | I               | Audio Left Channel Microphone Positive Input     | AVDD_AUD     | 15                   |
| AUD_MICCM2                            | I               | Audio Left Channel Microphone Negative Input     | AVDD_AUD     | 16                   |
| <b>USB 2.0 Interface</b>              |                 |                                                  |              |                      |
| USB2_DM                               | I/O             | USB 2.0 Differential Pair, Negative              | AVDD3P3_USB  | 102                  |
| USB2_DP                               | I/O             | USB 2.0 Differential Pair, Positive              | AVDD3P3_USB  | 101                  |
| <b>Power pins</b>                     |                 |                                                  |              |                      |
| VDD                                   | Core Power      | Digital Core Power                               |              | 23, 47, 74, 100, 118 |
| DVDD_DDR                              | Core Power      | Digital Power for DDR TX                         |              | 44                   |
| DVDD_DDR_RX                           | Core Power      | Digital Power for DDR RX LDO (0.1uF CAP to GND)  |              | 43                   |
| VDDIO_DATA                            | DDR Power       | IO Power for DDR Data                            |              | 46                   |
| VDDIO_MCLK                            | DDR Power       | IO Power for DDR Clock                           |              | 48                   |
| VDDIO_CMD                             | DDR Power       | IO Power for DDR Command                         |              | 50                   |
| AVDDIO_DRAM                           | DDR Power       | IO Power for embedded DRAM                       |              | 40, 49               |
| VDDP_1_3318                           | 1.8V/3.3V Power | Digital Input/Output Power for Domain 1          |              | 34                   |

| Signal Name | Signal Type     | Function                                | Power Domain | Pin Location |
|-------------|-----------------|-----------------------------------------|--------------|--------------|
| VDDP_2_3318 | 1.8V/3.3V Power | Digital Input/Output Power for Domain 2 |              | 64           |
| VDDP_3      | 3.3V Power      | Digital Input/Output Power for Domain 3 |              | 89           |
| VDDP_4      | 3.3V Power      | Digital Input/Output Power for Domain 4 |              | 97           |
| AVDD_PLL    | 3.3V Power      | Analog Power for PLL                    |              | 45           |
| AVDD_XTAL   | 3.3V Power      | Analog Power for XTAL                   |              | 126          |
| AVDD3P3_USB | 3.3V Power      | Analog Power for USB2.0                 |              | 103          |
| AVDD_ETH    | 3.3V Power      | Analog Power for Ethernet               |              | 104          |
| AVDD_AUD    | 3.3V Power      | Analog Power for Audio                  |              | 10           |
| AVDD_HD     | 3.3V Power      | Analog Power for Audio Head Phone       |              | 9            |
| GND_E       | GND             | Ground                                  |              | 62           |
| AVSS_HD     | GND             | Analog Ground for Audio Head Phone      |              | 6            |

4.3. Mechanical Dimensions

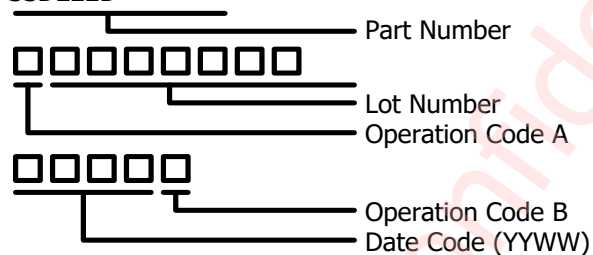


## 4.4. Ordering Guide

| Part Number | Ambient Temperature | Package Description | Package Option | RoHS Status    | Minimum Order Quantity |
|-------------|---------------------|---------------------|----------------|----------------|------------------------|
| SSD222D     | -20°C to +85°C      | QFN                 | 128-pin        | RoHS compliant | 1216ea                 |

## 4.5. Marking Information

SSD222D



## 5. ELECTRICAL SPECIFICATIONS

### 5.1. Interface Characteristics

| Parameter            | Symbol   | Min.                                     | Typ.   | Max.          | Unit |
|----------------------|----------|------------------------------------------|--------|---------------|------|
| DIGITAL INPUTS       |          |                                          |        |               |      |
| Input Voltage, High  | $V_{IH}$ | $V_{DDP} \times 0.7$ <small>Note</small> |        |               | V    |
| Input Voltage, Low   | $V_{IL}$ |                                          |        | 0.8           | V    |
| Input Current, High  | $I_{IH}$ |                                          |        | -1.0          | uA   |
| Input Current, Low   | $I_{IL}$ |                                          |        | 1.0           | uA   |
| Input Capacitance    |          |                                          | 5      |               | pF   |
| DIGITAL OUTPUTS      |          |                                          |        |               |      |
| Output Voltage, High | $V_{OH}$ | $V_{DDP} - 0.1$ <small>Note</small>      |        |               | V    |
| Output Voltage, Low  | $V_{OL}$ |                                          |        | 0.1           | V    |
| SAR ADC Input        |          | 0                                        |        | $V_{VDD\_33}$ | V    |
| AUDIO OUTPUTS        |          |                                          |        |               |      |
| Line-Out             |          |                                          | 2.54   |               | Vp-p |
| XTAL Specifications  |          |                                          |        |               |      |
| Input Voltage, High  | $V_{IH}$ | 2.0                                      |        | 3.6           | V    |
| Input Voltage, Low   | $V_{IL}$ | -0.3                                     |        | 0.8           | V    |
| Clock frequency      |          |                                          | 24     |               | MHz  |
| Crystal accuracy     |          |                                          | +/-30  |               | ppm  |
| Long-term jitter     |          |                                          | +/-500 |               | ps   |

**Note: VDDP typical voltage is 3.3V or 1.8V**

## 5.2. Recommended Operating Conditions

| Parameter                        | Symbol                | Min  | Typ. | Max. | Unit |
|----------------------------------|-----------------------|------|------|------|------|
| 3.3V Supply Voltage              | V <sub>VDD_33</sub>   | 3.14 | 3.3  | 3.46 | V    |
| 1.5V Supply Voltage (DDR III)    | V <sub>VDD_15</sub>   | 1.45 | 1.5  | 1.55 | V    |
| Core Power Supply Voltage (Core) | V <sub>VDD_core</sub> | 0.87 | 0.9  | 0.93 | V    |
| Ambient Temperature              | T <sub>A</sub>        | -20  |      | 85   | °C   |
| Junction Temperature             | T <sub>J</sub>        |      |      | 125  | °C   |

## 5.3. Absolute Maximum Ratings

| Parameter                        | Symbol                | Min  | Typ. | Max. | Unit |
|----------------------------------|-----------------------|------|------|------|------|
| 3.3V Supply Voltage              | V <sub>VDD_33</sub>   | 2.97 | 3.3  | 3.63 | V    |
| 1.5V Supply Voltage (DDR III)    | V <sub>VDD_15</sub>   |      |      | 1.65 | V    |
| Core Power Supply Voltage (Core) | V <sub>VDD_core</sub> |      |      | 1.1  | V    |
| Storage Temperature              | T <sub>STG</sub>      | -40  |      | 150  | °C   |

**Note:** Stresses above those listed in Absolute Maximum Ratings may cause permanent damage to the device. This is a stress rating only and does not imply functional operation of device. Exposure to absolute maximum ratings for extended periods may affect device reliability.

## 6. HARDWARE POWER SEQUENCE PROCEDURE

The timing requirements of the hardware reset signal are shown as below:

### 6.1. Hardware Reset

HWRESET: Chip Reset; High Reset (Level)

The HWRESET pin is suggested to connect with 3.3V standby as shown in Figure 1. The  $V_{IH}$  is 2V (Typ)  $\pm$  10% (2.2V~1.8V); the  $V_{IL}$  is 1.2V (Typ)  $\pm$  10% (1.08V~1.32V). The power sequence is as shown in Figure 2.

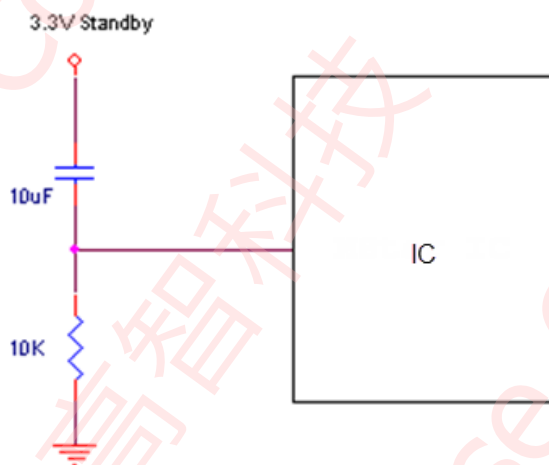
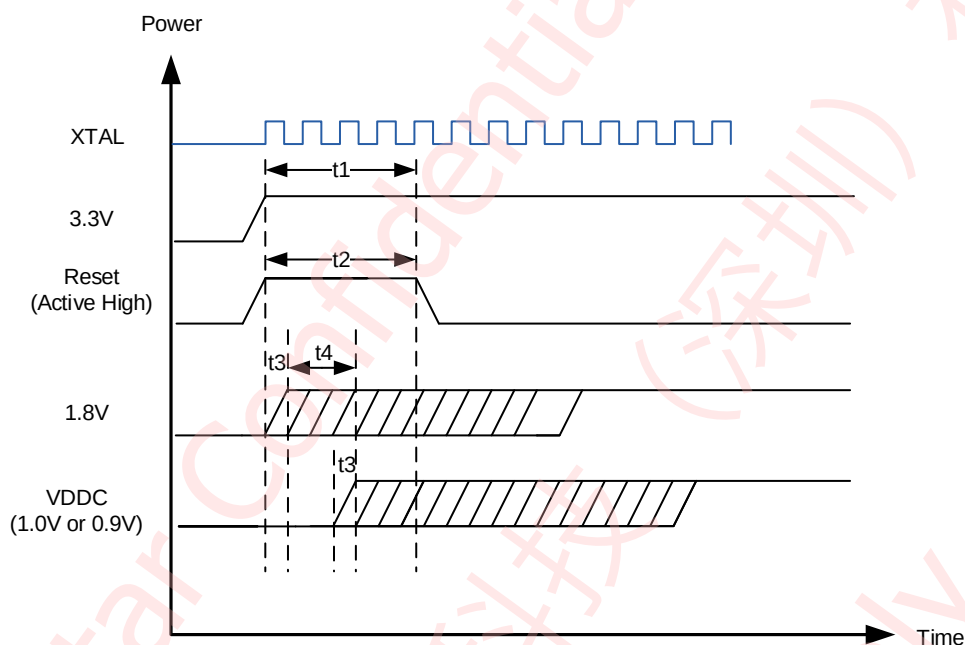


Figure 1: Reset Application Circuit

## 6.2. External Reset IC with External LDO

The timing is shown as Figure 2. The RST and power waveform must satisfy Figure 2 with parameters listed in Table 1.



Note:

\*3.3V (AVDD\_XTAL, AVDD\_ETH, AVDD3P3\_USB, AVDD\_PLL, AVDD\_AUD, AVDD\_HD, VDDP\_3, VDDP\_4, VDDP\_1\_3318 (if 3.3V is used), VDDP\_2\_3318 (if 3.3V is used))

\*1.0V/0.9V (VDD)

\*1.5V (AVDDIO\_DRAM, VDDIO\_DATA, VDDIO\_CMD, VDDIO\_MCLK)

\*3.3V/1.8V (VDDP\_1\_3318 (if 1.8V is used), VDDP\_2\_3318 (if 1.8V is used))

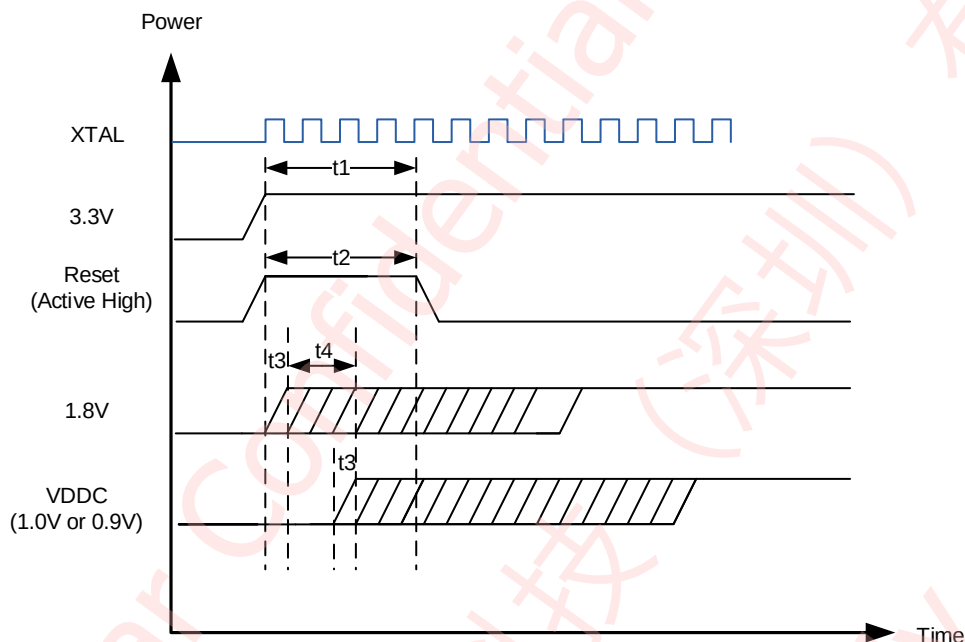
Figure 2: Power on Sequence

Table 1: Power Requirements

| Time           | Description                                                | Min | Typ. | Max | Unit |
|----------------|------------------------------------------------------------|-----|------|-----|------|
| t <sub>1</sub> | XTAL stable to Reset falling                               | 5   | —    | —   | ms   |
| t <sub>2</sub> | Reset pulse width                                          | 5   | —    | —   | ms   |
| t <sub>3</sub> | Normal 3.3V, 1.8V, and VDDC power rising time (0% to 100%) | —   | —    | 20  | ms   |
| t <sub>4</sub> | Normal 3.3V and 1.8V to VDDC lead time                     | 1   | —    | —   | ms   |

### 6.3. Without External Reset IC with External LDO

The timing is shown as Figure 3. The power waveform must satisfy Figure 3 with parameters listed in Table 2.



**Note:**

\*3.3V (AVDD\_XTAL, AVDD\_ETH, AVDD3P3\_USB, AVDD\_PLL, AVDD\_AUD, AVDD\_HD, VDDP\_3, VDDP\_4, VDDP\_1\_3318 (if 3.3V is used), VDDP\_2\_3318 (if 3.3V is used))

\*1.0V/0.9V (VDD)

\*1.5V (AVDDIO\_DRAM, VDDIO\_DATA, VDDIO\_CMD, VDDIO\_MCLK)

\*3.3V/1.8V (VDDP\_1\_3318 (if 1.8V is used), VDDP\_2\_3318 (if 1.8V is used))

Figure 3: Power on Sequence

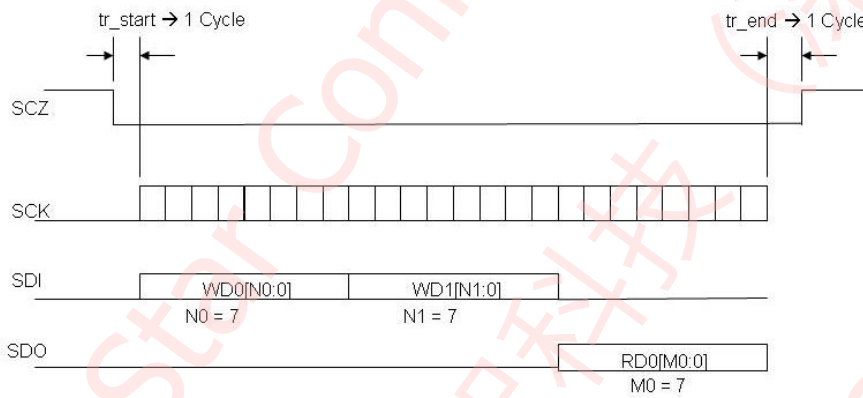
Table 2: Power Requirements

| Time           | Description                                                | Min | Typ. | Max | Unit |
|----------------|------------------------------------------------------------|-----|------|-----|------|
| t <sub>3</sub> | Normal 3.3V, 1.8V, and VDDC power rising time (0% to 100%) | —   | —    | 20  | ms   |
| t <sub>4</sub> | Normal 3.3V and 1.8V to VDDC lead time                     | 1   | —    | —   | ms   |

## 7. MSPI OPERATION EXAMPLE

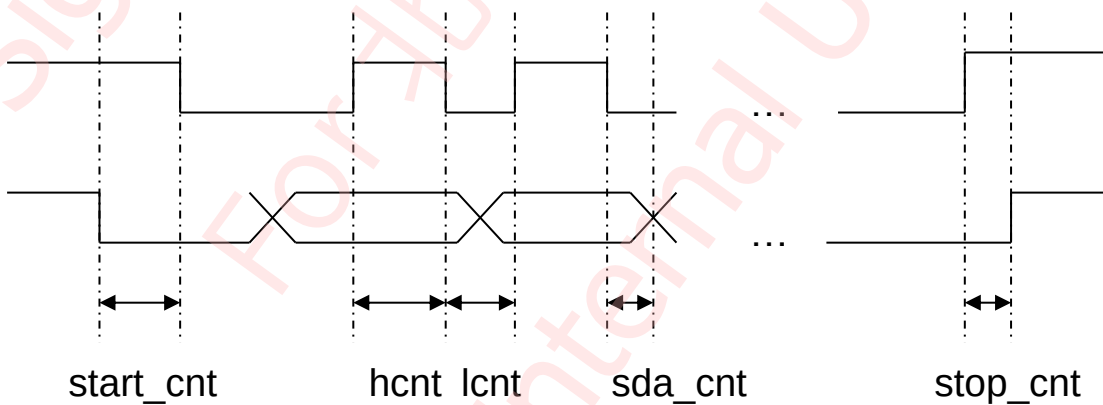
This section describes an example of MSPI operation.

- (0) Initialize
- (1) CS goes low
- (2) Write 2-Byte data
- (3) Read 1-Byte data
- (4) CS goes high



I2C clock frequency is configurable between 100KHz and 400KHz.

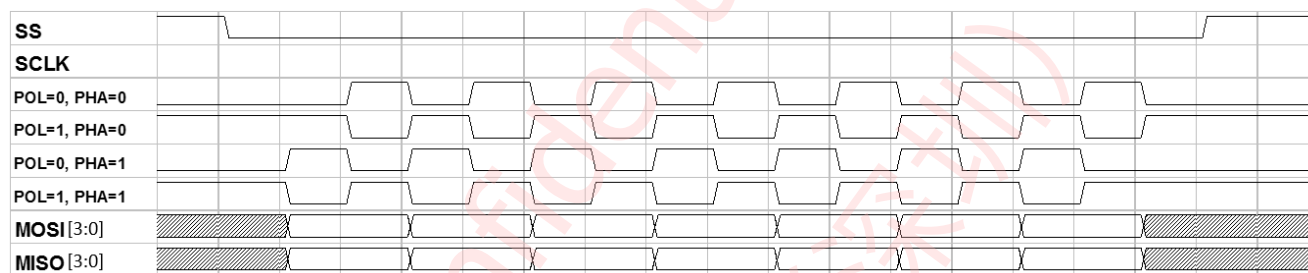
### 7.1. Set MIIC Speed



| clk_miic                        | 12MHz | 24MHz |
|---------------------------------|-------|-------|
| lcnt (>1.3us)                   | >16T  | >31T  |
| hcnt (>0.6us)                   | >8T   | >15T  |
| start (>0.6us)                  | >8T   | >15T  |
| stop (>0.6us)                   | >8T   | >15T  |
| between start and stop (>1.3us) | >16T  | >31T  |
| data_latch (>0us)               | >0T   | >0T   |
| sda change (<0.9us)             | <11T  | <22T  |

| Register Name    | Address    | Description                                           |
|------------------|------------|-------------------------------------------------------|
| reg_stop_cnt     | 'h08[15:0] | Sets the SCL and SDA count for stop                   |
| reg_hcnt         | 'h09[15:0] | Sets the SCL clock high-period count                  |
| reg_lcnt         | 'h0a[15:0] | Sets the SCL clock low-period count                   |
| reg_sda_cnt      | 'h0b[15:0] | Sets the clock count between falling edge SCL and SDA |
| reg_start_cnt    | 'h0c[15:0] | Sets the SCL and SDA count for start                  |
| reg_data_lat_cnt | 'h0d[15:0] | Sets the data latch timing                            |

## 8. PSPI INTERFACE

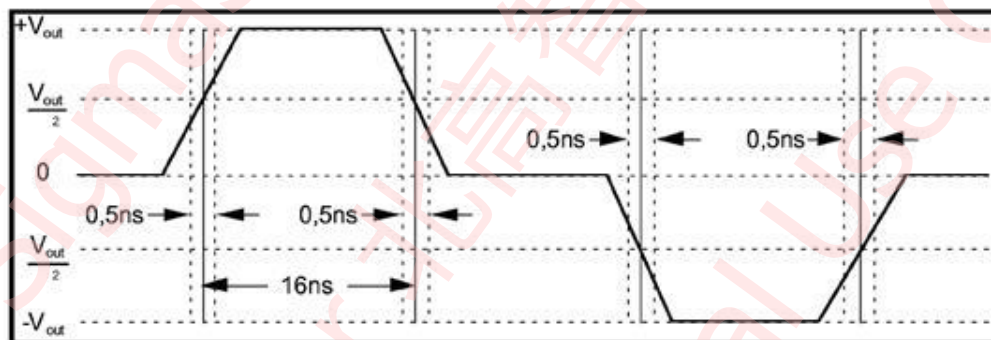


| POL | PHA | Transfer                                         |
|-----|-----|--------------------------------------------------|
| 0   | 0   | Receive at rising edge, transmit at falling edge |
| 1   | 0   | Receive at falling edge, transmit at rising edge |
| 0   | 1   | Receive at falling edge, transmit at rising edge |
| 1   | 1   | Receive at rising edge, transmit at falling edge |

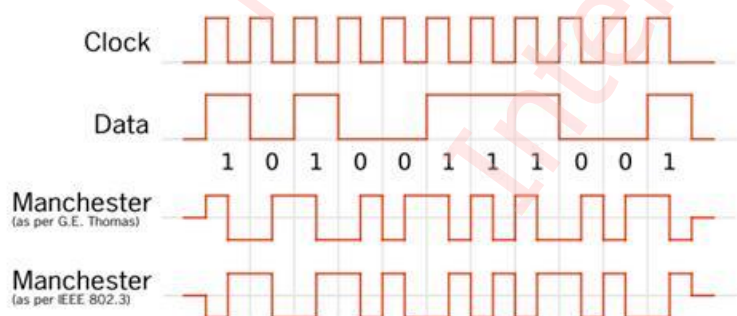
## 9. EPHY INTERFACE

| Parameter                              | Min   | Typ | Max  | Unit              |
|----------------------------------------|-------|-----|------|-------------------|
| ETHERNET ANALOG INTERFACE (10BASE-T)   |       |     |      |                   |
| Analog Input Range                     | 4.4   | 5   | 5.6  | V <sub>dp-p</sub> |
| Differential Input Impedance           |       | 100 |      | ohm               |
| ETHERNET ANALOG INTERFACE (100BASE-TX) |       |     |      |                   |
| Analog Input Range                     | 1.9   | 2   | 2.1  | V <sub>dp-p</sub> |
| Differential Input Impedance           |       | 100 |      | ohm               |
| Rise/Fall Time                         | 3     | 4   | 5    | ns                |
| Rise/Fall Time Symmetry                |       |     | 0.5  | ns                |
| Duty Cycle Distortion                  | -0.25 |     | 0.25 | ns                |
| Amplitude Symmetry                     | 98    | 100 | 102  | %                 |
| Overshoot                              |       |     | 5    | %                 |

### 9.1. 100BASE-TX



### 9.2. 10BASE-T



## 10. THERMAL RESISTANCE (°C/W)

### 10.1. Thermal simulation mode

1. PCB conditions (JEDEC JESD51-5)
2. PCB layers: 2L (1S1P)
3. PCB dimensions (mm x mm): 76.2 x 114.3
4. PCB thickness (mm): 1.6

| PKG Type | PKG Size (mm) /<br>Pin Count | PCB Layer | Theta jc (C/W) | Theta jb (C/W) | Ta (C) | Tj (C) | Theta ja (C/W) |
|----------|------------------------------|-----------|----------------|----------------|--------|--------|----------------|
|          |                              |           |                |                |        |        | 0 m/s          |
| QFN      | 12.3x12.3 / 128L             | 2L        | 6.9            | 10.82          | 75     | 125    | 22.6           |